

Verdant Coil – Game Design Document

■ Concept Summary

Verdant Coil is an asynchronous, tile-based dungeon builder and explorer game set in a surreal, biotech-inspired world. Players design and publish living maze-like dungeons called Coils, filled with hazardous, evolving organic structures. Others explore these Coils by guiding a customizable creature (a Crawler) through the maze to the Heartroot.

■ Core Gameplay Loop

1. Explore Mode (PvE)

- Control a Crawler in a turn-based, top-down grid world.
- Traverse living mazes built by other players.
- Encounter and adapt to hazards, terrain, and puzzle-like obstacles.
- Earn Nutrient (primary resource) by completing levels.

2. Build Mode (Creator)

- Design organic maze-like levels using a modular tile palette.
- Levels must be solvable and validated by the builder before publishing.
- Gain Sporeprint (secondary resource) when others complete your Coils.

3. Progression

- Nutrient: Used to expand your crawler or unlock builder modules.
- Sporeprint: Used to unlock new tiles or increase biomass budgets.

■■■ Player Unit – The Crawler

Crawlers are turn-based units with customizable abilities and stats, including: • Health • Armor • Speed • Traction • Perception • Acid Reserve • Toggleable Upgrades like Hardened Skin, Acid Sac, Photoreceptors, etc.

■ Level Structure & Tile Types

• Grid size: 24×24 default, expandable to 48×48 • Biomass budget limits tile count • Start point: Bottom center by default • Goal: Heartroot tile must be reachable • One Heartroot per level; must be solvable

Tile Categories

• Structural Tiles: Flesh Tile, Digest Wall, Pulse Gate, Heartroot • Hazard Tiles: Acid Pool, Sticky Web, Quivering Spores • Functional Tiles: Trigger Node, Membrane, Camouflage • Projectile Tiles: Guard Nodule, Spine Shooter, Tendril Lash

■■ Runtime Mechanics

Turn-based logic, in order: 1. Player moves 2. AOE effects trigger 3. Traps activate 4. Projectiles update 5. Enemies act (if any)

■ User Interface

Explorer Mode UI

• Top bar: Nutrient, Sporeprint, Active Upgrades • Bottom corners: Restart and Menu buttons • Keyboard hotkeys to toggle upgrades

Builder Mode UI

• Left sidebar: Tile palette • Top bar: Biomass usage • Bottom bar: Validate, Save, Publish • Highlight illegal placements and show tooltips on hover

■ Visual Style

• Aesthetic: Alien biotech with glowing greens, teals, and bone whites • Fonts: Rounded sans-serif • UI: Translucent panels, soft edges, modular CanvasLayer-based system

■ MVP Scope

- Grid-based crawler with upgrades • Basic level builder with tile placement and validation • JSON-based save/load system • Local-only; backend optional for post-MVP

■ *Asynchronous System (Planned)*

- Online backend for player data and levels • Server-tracked completions and anti-abuse logic • Reputation systems, analytics, and featured content

■ Design Pillars

- Asynchronous Creative Ecosystem • Biotech World-Building • Progress Through Participation • Tactical Depth + Puzzle Solving